Integrals

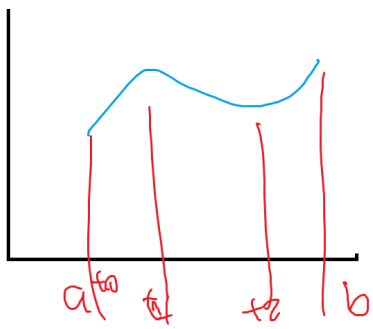
ICPR

January 25, 2022

1 The idea of integration

Assume $f(x) > 0$

We want the area under the curve of $f(x)$ on $[a,\ b]$. Split the area underneath into many rectangles and add up their total area.



Partition: the way we pick the bases of the rectangles. There's nothing stopping you from splitting the partitions into equal length intervals, but you should. For example, Partition P: some number of rectangles between a and b. You can refine a partition by adding other points. The result would be called a **refinement.**

Choosing the heights of the rectangles: There are two methods we can do (WIDTH OF RECTANGLE IS CONTROLLED HERE):

- Upper sum: Highest point; maximum value of f (supremum) of f in the rectangle; called $U_f(P)$; area over-estimator $\geq$

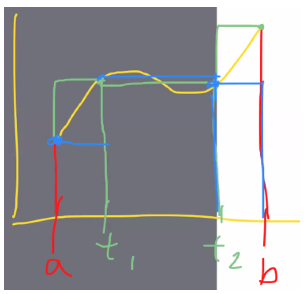
- Lower sum: Lowest point; takes the infimum of f; called $L_f(P)$; area under-estimator

While these two are great for theoretical purposes they aren't computationally feasible. Here are the methods that are easy for computation:

- Right-point method

- Left-point method

There are drawbacks of these two methods (often due to inaccuracies), but the height is already determined for you the moment you choose where your rectangles go. For example:



$$h_1 = f(t_1),\ h_2 = f(t_2),\ h_3 = f(t_3)$$

1.1 Formulas for these methods

Given a partition of $[a,\ b]$ where $t_0 = a$ and $t_n = b$

$$P = t_0 < t_1 < ... < t_n$$

We may compute the total area explicitly. In the ith rectangle with base $[t_{i-1},\ t_i]$, the right most endpoint is t_i and left is t_{i-1}.

Total area by the right-point method:

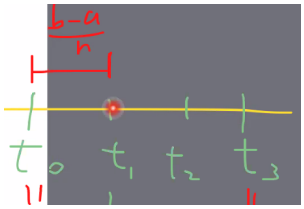
$$A = \sum_{i=1}^{n} b_i \cdot h_i = \sum_{i=1}^{n} (t_i - t_{i-1}) \cdot f(t_i)$$

For the left-point method:

$$A = \sum_{i=1}^{i} b_i \cdot h_i = \sum_{i=1}^{n} (t_i - t_{i-1}) \cdot f(t_{i-1})$$

The right-point method is slightly easier.

1.2 Even splits



If we split an interval $[a,\ b]$ into n-equal sized pieces:

Each rectangle would have base length $b_i = \frac{b-a}{n}$

This means to determine where a split point t is, where $t_0 = a$:

$$t_1 = a + \frac{b-a}{n}$$
$$t_i = a + i\frac{(b-a)}{n}$$

Using equal-length intervals, using rectangle width and height (depends on right point or left point):

$$A_R = \sum_{i=1}^{n} b_i \cdot h_i = \sum_{i=1}^{n} \frac{b-a}{n} \cdot f\left(a + i\frac{(b-a)}{n}\right)$$
$$A_L = \sum_{i=1}^{n} b_i \cdot h_i = \sum_{i=1}^{n} \frac{b-a}{n} \cdot f\left(a + (i-1)\frac{(b-a)}{n}\right)$$

1.3 True area

As we use more rectangles, the better approximation to the **true area** under the curve we get of $f(x)$.

An informal definition of the true area (named integral) under the curve of $f(x)$ is approximated using more and more angles.

$$A_T = \int_a^b f(x)\mathrm{d}x = \lim_{n\to\infty} \sum_{i=1}^{n} b_i \cdot h_i$$

(Remember how b_i and h_i are defined, but h_i represents some way we choose the height of the rectangles. As the number of rectangles reaches infinity, we are supposed to get the true answer every time.)

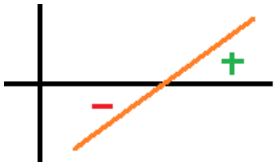
When n becomes large, b_i becomes verry narrow so it is centered around a point x, so we can change that notation from $b_i \to dx$. $f(x)$ comes from the fact that $h_i \to f(x)$ when b_i, the box length becomes very narrow.

1.4 Positive and negative area

About net area:

When $f(x) > 0$, we count the area above the x-axis and below $f(x)$ to be positive area.

When $f(x) < 0$, we count the area below the x-axis and below $f(x)$ to be negative area.



The net area is the sum of both positive area and negative area.

1.5 Integrals without integral rules

Compute $\int_0^1 2x dx$. Area is $A = \frac{1}{2} \cdot 1 \cdot 2 = 1$ (triangle).

Using integrals, we can do this instead, for $f(x) = 2x,\ [a = \ 0, b = 1]$:

$$\begin{aligned} A &= \sum_{i=1}^{n} b_i \cdot h_i = \sum_{i=1}^{n} \frac{b-a}{n} \cdot f\left(a + i\frac{(b-a)}{n}\right) \\ &= \sum_{i=1}^{n} \frac{1}{n} \cdot f\left(i\frac{1}{n}\right) = \sum_{i=1}^{n} \frac{1}{n} \cdot 2\left(i\frac{1}{n}\right) \\ &= 2\sum_{i=1}^{n} \frac{i}{n^2} = \frac{2}{n^2}\sum_{i=1}^{n} i = \frac{2}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{n} \end{aligned}$$

Anything that does not involve i can be taken out of summations, so you can use this fact to your advantage.

Taking the limit of n to infinity:

$$\lim_{n\to\infty} \frac{n+1}{n} = \frac{n}{n} + \frac{1}{n} = 1$$

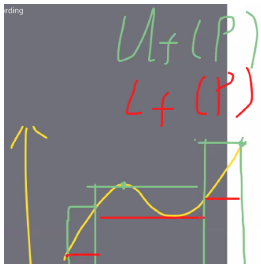
2 1D Integration

The upper and lower sums are the methods to theoretically find the integral. The true area is always sandwiched between the upper sum and the lower sum.

For any partitions P_1 and P_2:

$$L_f(P_1) \leq \text{True Area} \leq U_f(P_2)$$

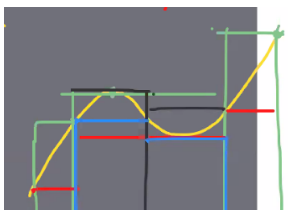
This holds even if P_1 and P_2 are different.



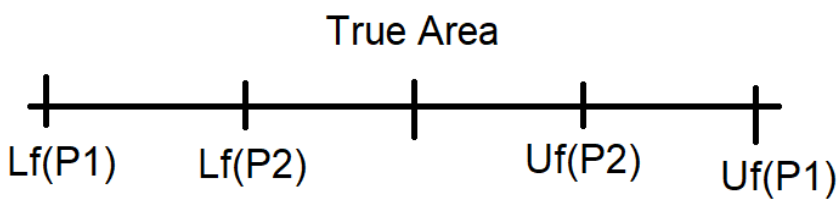
By picking more rectangles, we can make the upper sum and lower sum closer and closer together.

If P_2 is a **refinement** of P_1 (P_2 has all the rectangles of P_1 and some more), then P_2 should provide a better estimate:

$$L_f(P_1) \leq L_f(P_2) \leq \text{True Area} \leq U_f(P_2) \leq U_f(P_1)$$



Here, there are 4 rectangles instead of the 3. The area of the green rectangles is larger than the area of the black rectangles, and the area of the red rectangles is less than the area of the blue rectangles.



The more rectangles, the closer to the true area we are, but we haven't defined true area yet.

2.1 Defining true area without defining true area

Approximate the true area up to some error ε we are comfortable with, such that $U_f(P) - L_f(P) < \varepsilon$. Pick an ε as small as possible.

Definition 1. *A function is* ***integrable*** *if the upper sum and lower sum can be made as close as you want (by picking more rectangles).*

$$\forall \varepsilon > 0,\ \exists P \text{ such that } U_f(P) - L_f(P) < \varepsilon$$

Note the same partition P is used here in the upper and lower sum. The theory here is that the error can be made as small as possible (you can always make your error smaller provided that it is positive), but there will always be some error.

The **true area** should be the minimum of all possible upper sums whilst also being equal to the maximum of all possible lower sums. There are however infinitely as many upper and lower sums, so a maximum and minimum wouldn't exist. We could believe that an intuitive max/min exists for them:

- The supremum of a set S is intuitively understood as the maximum of the set S.

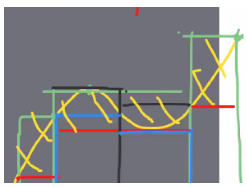
- The infimum of a set S is intuitively understood as the minimum of the set S.
- This is from point 13 of the axioms of real numbers.

Definition 2. *True area:*

$$\begin{aligned} \textit{True Area} = \int_a^b f &= \textit{infimum of all possible } \left\{U_f(P)\right\} = \overline{I}_a^b \\ &= \textit{supremum of all possible } \left\{L_f(P)\right\} \end{aligned}$$

I'd rather use definition 1.

$U_f(P) - L_f(P)$ is liked the shaded area below:



Almost all functions are integrable. Even though definition 1 can be used, it doesn't define true area.

2.2 Proving that a function is integrable

Show whether the following functions are integrable.

$$f(x) = 2,\ x \in [0,\ 2]$$

Prove $\forall \varepsilon > 0,\ \exists P,\ U_f(P) - L_f(P) < \varepsilon$.

Proof. Let $\varepsilon > 0$. Pick P, which is the rectangle from 0 to 2. There is only one rectangle, the base being 2 and the height being 2 by looking at the function, so $U_f(P) = 2_{\text{height}} \cdot 2_{\text{base}} = 4$.

For $L_f(P)$, calculate it by $L_f(P) = 2_{\text{height}} \cdot 2_{\text{base}} = 4$.

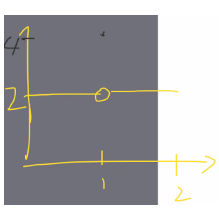
Then:

$$U_f(P) - L_f(P) = 2 - 2 = 0 < \varepsilon$$

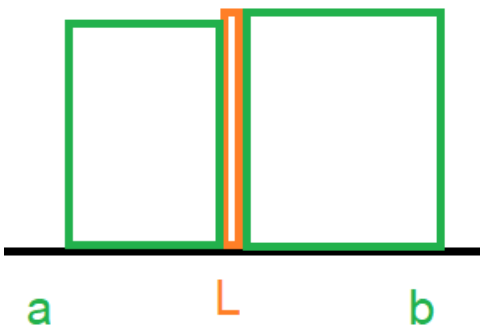
Here, the error is always less than epsilon. ■

We can try something more complicated:

$$f(x) = \begin{cases} 2 & x \neq 1 \\ 4 & x = 1 \end{cases},\ x \in [0,\ 2]$$



This is a jump discontinuity.



Proof. Let $\varepsilon > 0$. Pick partition $P : 0 \to 1 - \frac{l}{2} \to 1 + \frac{l}{2} \to 2$ (the partitions do not need to be equal length).

Effectively, you are creating three rectangles. This means:

$$\begin{aligned} U_f(P) &= \overbrace{b_1 \cdot 2}^{\text{left rectangle}} + \overbrace{l \cdot 4}^{\text{middle rectangle}} + \overbrace{b_3 \cdot 2}^{\text{right rectangle}} \\ L_f(P) &= b_1 \cdot 2 + l \cdot 2 + b_2 \cdot 2 \end{aligned}$$

Now:

$$\begin{aligned} U_f(P) - L_f(P) &< \varepsilon \\ (2b_1 + 4l + 2b_3) - (2b_1 + 2l + 2b_2) &< \varepsilon \end{aligned}$$

Choose $l = \frac{\varepsilon}{4}$. You can choose whatever you want, but you should be choosing l such that the end result does not look ugly.

$$\begin{aligned} 2b_1 + \varepsilon + 2b_3 - 2b_1 - \frac{\varepsilon}{2} - 2b_2 &< \varepsilon \\ \frac{\varepsilon}{2} &< \varepsilon \end{aligned}$$

■

What does this mean? Integrals are more versatile than the derivative. It can handle discontinuities, which can be done by picking small rectangles around discontinuities, which will control the discontinuity for you.

Another example:

$$f(x) = \begin{cases} 2 & x \neq 1 \\ 4 & x = 1 \end{cases}, \ x \in [0,\ 2]$$

Proof. Let $\varepsilon > 0$, and pick $P : 0 \to 1 - \frac{l}{2} \to 1 + \frac{l}{2} \to 2 - \frac{l}{2} \to 2$. This means

$$\begin{aligned} U_f(P) &= 2b_1 + 4l + 4b_3 + 6 \cdot \frac{l}{2} \\ L_f(P) &= 2b_1 + 2l + 4b_3 + 4 \cdot \frac{l}{2} \\ U_f(P) - L_f(P) &< \varepsilon \\ 2b_1 + 4l + 4b_3 + 6 \cdot \frac{l}{2} - 2b_1 - 2l - 4b_3 - 4 \cdot \frac{l}{2} &< \varepsilon \\ (4-2)l + \frac{(6-4)l}{2} = 3l &< \varepsilon \end{aligned}$$

Pick $l = \frac{\varepsilon}{6}$

$$\begin{aligned} 3 \cdot \frac{\varepsilon}{6} &< \varepsilon \\ \frac{\varepsilon}{2} &< \varepsilon \end{aligned}$$

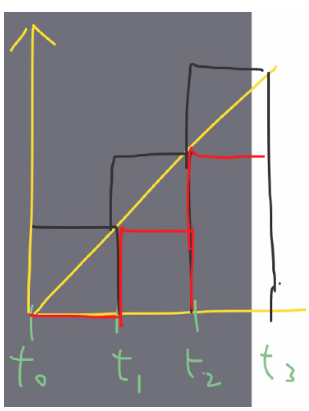
■

This method handles all discontinuities.

For a function that isn't flat: $f(x) = x,\ x \in [0,\ 2]$

$$\begin{aligned} b_1 h_1 &= b_2 h_2 \\ U_f(P) &= 1 \cdot 1 + 1 \cdot 2 \\ L_f(P) &= 1 \cdot 2 + 1 \cdot 1 \end{aligned}$$

Notice that f is increasing so if you pick the same base for each rectangle, then the adjacent rectangle's area of the upper sum and the lower sum would cancel.



Proof. Let $\varepsilon > 0$. Let P be n-equal spaced intervals, meaning $b = \frac{2}{n}$. Pick n such that $\frac{4}{n} < \varepsilon$.

Then:

$$U_f(P) = b \cdot f(t_1) + b \cdot f(t_2) + \cdots + b \cdot f(t_n)$$
$$L_f(P) = b \cdot f(t_0) + b \cdot f(t_1) + \cdots + b \cdot f(t_{n-1})$$

$$U_f(P) - L_f(P) = b \cdot f(t_n) - b \cdot f(t_0) = b\left(f(t_n) - f(t_0)\right) = \frac{2}{n} \cdot 2 = \frac{4}{n} < \varepsilon$$

■

2.2.1 Finitely many discontinuities

Suppose $g(x)$ is integrable and bounded (by M) on $[a,\ b]$. Let $f(x) = g(x)$ except at finitely many points $\{c_1,\ \ldots,\ c_n\}$ where $f(c_i) \neq g(c_i)$.

The idea is to have small rectangles around each of the discontinuities. Make each of the errors $\frac{\varepsilon}{2n}$ and the total error is $n \cdot \frac{\varepsilon}{2n} = \frac{\varepsilon}{2} < \varepsilon$.

Key word: **finitely** as many discontinuities.

Basically, if you have a continuous function with finitely many holes.

2.2.2 Infinitely as many discontinuities

This will cause you to set something like $\frac{\varepsilon}{\infty} = 0$, which can't happen, as the error can't be zero. The techniques of splitting up partitions and creating rectangles on discontinuities wouldn't work.

Example, where $\mathbb{Q}$ (standing for quotient) means rational numbers, which can be represented as $\frac{p}{q},\ p,\ q \in \mathbb{Z}$:

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

There are a lot of rational numbers (more like infinitely as many) scattered between $[0,\ 1]$. It is an infinitely scattering of points in $[0,\ 1]$. It is not a solid line; irrational numbers like $\frac{\pi}{4}$ exist. This means $f\left(\frac{\pi}{4}\right) = 0$. But also $\frac{\pi}{8}$ is irrational, so $f\left(\frac{\pi}{8}\right) = 0$, and so is $\frac{\pi}{16}$, $\frac{\pi}{5}$, and so on, and don't forget $\frac{e}{5}$, $\sqrt{2}$, and so on. There is an infinite scattering of points at $y = 0$, as there are a lot of irrational numbers scattered at $y = 0$.

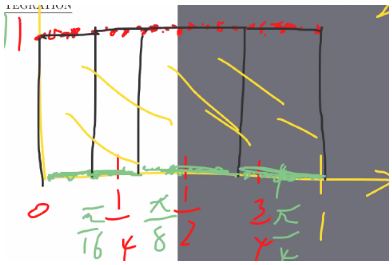
At a visual standpoint, there are points at $y = 1$ infinitely scattered and points at $y = 0$ infinitely scattered.

The trick of controlling errors can only work for finite discontinuities. If there are infinitely as many discontinuities, it feels like the function is not continuous everywhere, or even anywhere. At every single point, there's a discontinuity.

It does not mean that the function isn't integrable, but we run into this problem (note that $a < b$):

$$\forall [a,\ b],\ \exists x_1 \in \mathbb{Q},\ x_1 \in [a,\ b]$$
$$\forall [a,\ b],\ \exists x_1 \notin \mathbb{Q},\ x_2 \in [a,\ b]$$

We can see where the problem comes.



The upper sum for any interval (that isn't squished to a point) will always be 1, and 0 for the lower sum. There will always be an error of 1 overall, and $1 \not< \varepsilon$. Is it possible to make the error less than ε? Regardless of the partition you choose, the error will stay at 1.

What does it mean to be integrable? $\forall \varepsilon > 0,\ \exists P,\ U_f(P) - L_f(P) < \varepsilon$. I don't think this can be satisfied.

What is the negation of this definition?

$$\exists \varepsilon > 0,\ \forall P,\ U_f(P) - L_f(P) \geq \varepsilon$$

This is our proof that $f(x)$ here isn't integrable.

Proof. Pick $\varepsilon = \frac{1}{2}$. Let P be arbitrary. Regardless of how we choose our partition P, $U_f(P) - L_f(P)$ will always be greater or equal than ε:

$$\begin{aligned} U_f(P) &= \sum b_i \cdot h_i = \sum b_i \cdot 1 = 1 \\ L_f(P) &= \sum b_i \cdot h_i = \sum b_i \cdot 0 = 0 \end{aligned}$$

This always happens. What can we conclude?

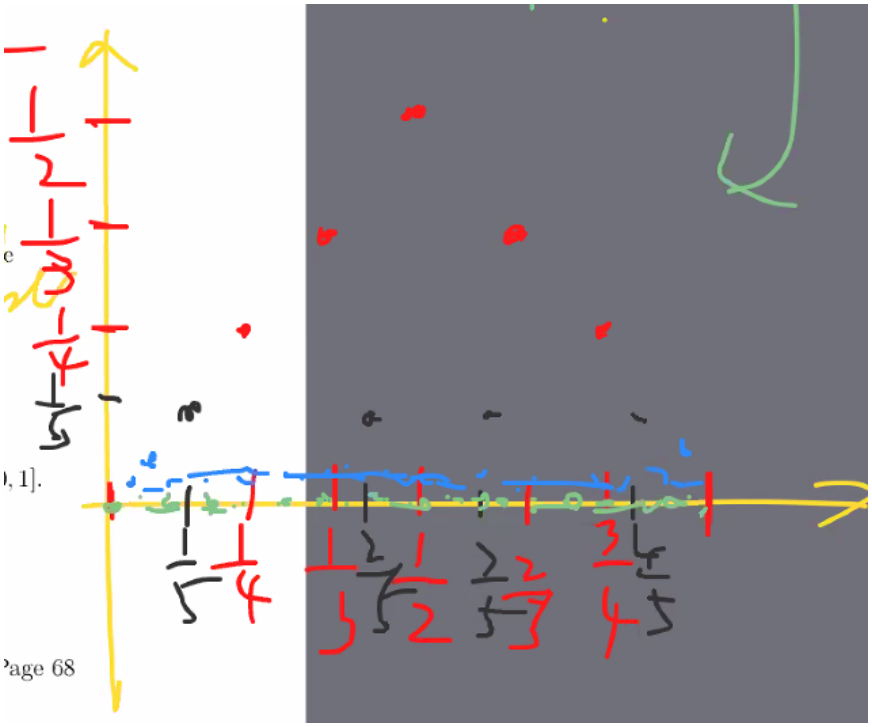
$$U_f(P) - L_f(P) = 1$$

...regardless of what P we choose. The error here is always going to be greater than 1, greater than or equal to the ε we set, $\frac{1}{2}$. We are unable to minimize the error to be under ε. ■

Ever so slightly harder example:

$$f(x) = \begin{cases} \frac{1}{q} & x = \frac{p}{q} \text{ written as an irreducible fraction} \\ 0 & x \notin \mathbb{Q} \end{cases} \quad x \in [0,\ 1]$$

What does it mean? $f\left(\frac{2}{3}\right) = \frac{1}{3},\ f\left(\frac{4}{5}\right) = \frac{1}{5}, f\left(\frac{6}{10}\right) = \frac{1}{5}$



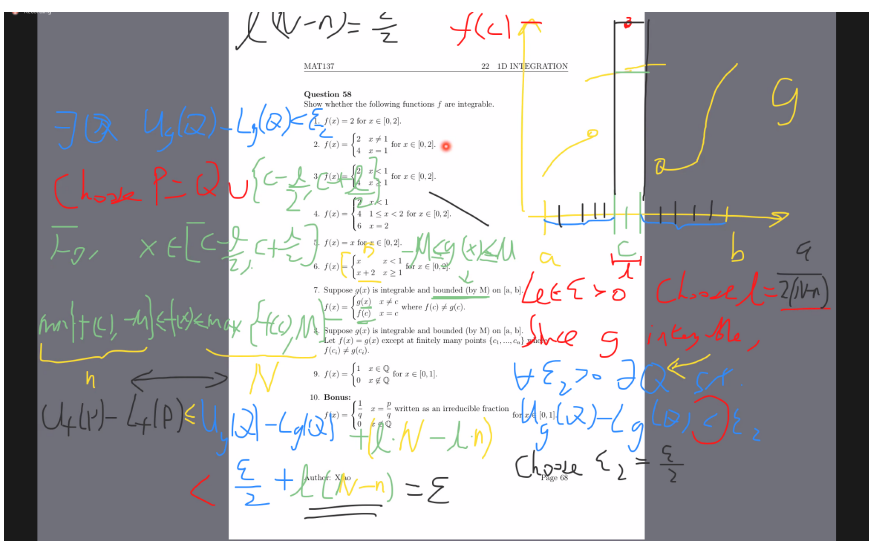
2.2.3 That for an increasing function

Suppose f is increasing on $[a,\ b]$. Prove f is integrable. Do not assume f is continuous.

Definition 3. *A function which is increasing or decreasing would be called monotonic.*

2.2.4 Debunking Q7

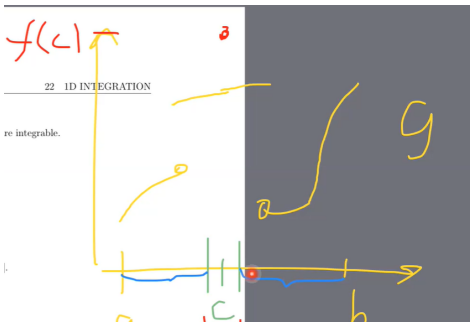
Suppose $g(x)$ is integrable and bounded by M on $[a,\ b]$



2.2.5 Debunking Question 6

Suppose $g(x)$ is integrable and bounded (by M) on $[a,\ b]$.

$$f(x) = \begin{cases} g(x) & \text{if } x \neq c \\ f(c) & \text{if } x = c \end{cases}, \text{ where } f(c) \neq g(c)$$



Proof. Let $\varepsilon > 0$. Choose $l = \frac{\varepsilon}{2(N-n)}$ (technically, we were supposed to define N and n earlier).

Since g is integrable, $\forall \varepsilon_2 > 0,\ \exists Q,$ such that $U_g(Q) - L_g(Q) < \varepsilon_2$ (there exists partition Q such that $U_g(Q) - L_g(Q)$.

Because g is integrable, we can remove parts of the partition in parts where we don't need it, such as the l-length region around c. This means the partition Q is only in the blue region.

Choose $\varepsilon_2 = \frac{\varepsilon}{2}$. This means the following:

- The error in the blue region must be at most $\frac{\varepsilon}{4}$.
- The error in the green region must be at most $\frac{\varepsilon}{4}$.
- The total error we will be making will be $\frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}$.

$$\exists Q,\ U_g(Q) - L_g(Q) < \varepsilon_2$$

Choose $P = Q \cup \left\{c - \frac{l}{2},\ c + \frac{l}{2}\right\}$ (Q union the green rectangle). There is some way to pick rectangles in the partition of Q such that the total error in the blue region will be less than $\frac{\varepsilon}{2}$.

Focusing on the error in the green region (and we must see how much error at most there can be):

For $x \in \left[c - \frac{l}{2},\ c + \frac{l}{2}\right]$, what can be the error at most? It should be less than ε. Here, f can be at most $f(c)$, but g could be even larger, so

$$\begin{aligned} x &\leq \max\left(f(x), M\right) \\ \min\left(f(x),\ -M\right) &\leq x \end{aligned}$$

This is based on the bounds g, as the function g may not go higher than M or lower than $-M$. Also, here are our bounds for x if it is in $\left[c - \frac{l}{2},\ c + \frac{l}{2}\right]$.

$$\min\left(f(c),\ -M\right) \leq f(x) \leq \max\left(f(c), M\right)$$

These two values are simply two numbers, so let $n = \min\left(f(c),\ -M\right)$ and $N = \max\left(f(c),\ M\right)$.

So

$$U_f(P) - L_f(P) = \overbrace{U_g(Q) - L_g(Q)}^{\text{blue region error}} + \overbrace{l \cdot N - l \cdot n}^{\text{green region error}}$$
$$= \frac{\varepsilon}{4} + \overbrace{l(N-n)}^{\text{which we pick to be } \frac{\varepsilon}{4}} = \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}$$

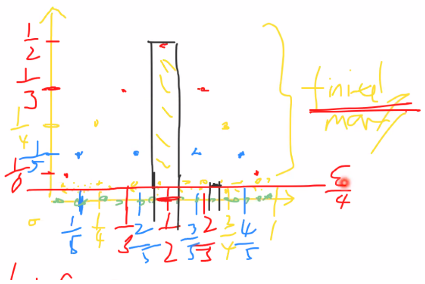
■

I don't get it. Our goal is to prove that f is integrable. This means we must prove that $\forall \varepsilon > 0,\ \exists P,\ U_f(P) - L_f(P) < \varepsilon$, knowing that we have g and the fact that it is integrable. Here's how we did it:

1. Name g's partition Q, which is actually from a to $c - \frac{l}{2}$, and then there is a region where we are not in partition Q, and then partition Q resumes at $c + \frac{l}{2}$ all the way to b. Yes, partitions can be split up.
2. Because we know g is integrable, we know the error it must not exceed: ε_2, which is universally quantified. We can let it be anything, so use that opportunity to assign it to something like $\frac{\varepsilon}{9999}$.
3. Choose the partition of f, which is the function we wish to prove integrable in the interval $[a,\ b]$. Pick it to be partition $Q \cup$ the gap in the middle, which is $\left\{c - \frac{l}{2},\ c + \frac{l}{2}\right\}$ – just one rectangle, and we have made gaps like this before because we want l to be the length of that rectangle.
4. This still concerns the partition of f we've just chosen. We know that $\forall x \in$ the gap $\left\{c - \frac{l}{2},\ c + \frac{l}{2}\right\}$, x can't:
   a. Go higher than $f(c)$ or M. We only worry about $f(c)$ when $x = c$, so M covers all the other cases: the maximum possible value of g.
   b. Go lower than $f(c)$ or $-M$. We only worry about $f(c)$ when $x = c$, so $-M$ covers all the other cases: the minimum possible value of g.
   c. We end up with this expression: $\min(f(c),\ -M) \leq f(x) \leq \max(f(c),\ M)$ We give aliases: $n = \min(f(c),\ -M)$ and $N = \max(f(c),\ M)$ although we totally didn't have to. The reason why we chose M and $-M$ instead of some $g(x)$ thing is because M and $-M$ are constants and will always be an overestimate/underestimate of where g always is.
5. We conclude that the error caused by that rectangle is $\leq (l \cdot N - l \cdot n)$. $l \cdot N$ is the upper height bound of that rectangle, and $l \cdot n$ is the lower height bound of the rectangle. A reminder that l is the length of the rectangle, and remember that we overestimated the upper bound and underestimated the lower bound because instead of choosing $g(x)$, we just looked at the highest/lowest possible value of g.
6. We return to the partition P we've chosen because before after acquiring more information about the rectangle in the middle in the region $\left\{c - \frac{l}{2},\ c + \frac{l}{2}\right\}$. Because we overestimated the error bound, we can safely say that $U_f(P) - L_f(P) \leq U_g(Q) - L_g(Q) + (l \cdot N - l \cdot\ n)$ and can finally choose $U_g(Q) - L_g(Q) < \frac{\varepsilon}{9999999}$ and choose l such that $(l \cdot N - l \cdot n) < \frac{\varepsilon}{9999999}$ in the front of the proof and finally prove that $U_f(P) - L_f(P) < \varepsilon$.

2.2.6 Debunking question 10

$$f(x) = \begin{cases} \frac{1}{q} & x = \frac{p}{q} \text{ written as an irreducible fraction} \\ 0 & x \notin \mathbb{Q} \end{cases} \qquad x \in [0,\ 1]$$



Let $\varepsilon > 0$. There are finitely many x with $f(x) > \frac{\varepsilon}{4}$. There is a list of possible x for which $f(x) > \frac{\varepsilon}{4}$.

$$x : \{c_1,\ c_2,\ \ldots,\ c_n\} = \left\{\frac{1}{2}, \frac{1}{3}, \frac{2}{3},\ \ldots\right\}$$

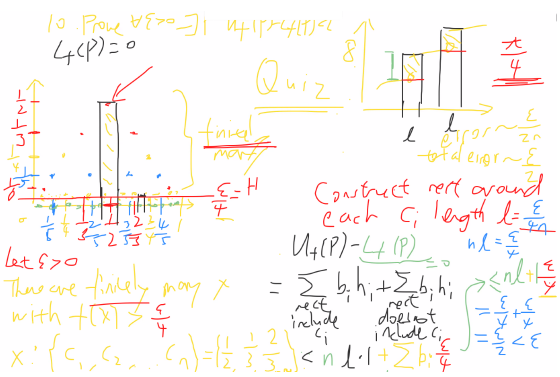
Well, excluding duplicates.

Construct rectangles around each c_i length $l = \frac{\varepsilon}{4n}$ (we chose that later).

$$U_f(P) - L_f(P) = \sum_{\substack{\text{rectangles}\\ \text{including } c_i}} b_i h_i + \sum_{\substack{\text{rectangles not}\\ \text{including } c_i}} b_i h_i$$

In the first (left) sum, there are n rectangles with the base being $n \cdot l$ with the height values all being less than $\frac{1}{2}$ (which implies they are all less than 1). There are n rectangles at most, where each base is chosen to have length l, and the height of each rectangle which includes one of these c_i points:

$$\sum_{\substack{\text{rectangles}\\ \text{including } c_i}} b_i h_i + \sum_{\substack{\text{rectangles not}\\ \text{including } c_i}} b_i h_i$$

What about the sum on the right? These rectangles do not include any of the $c_{1\ldots i\ldots n}$, so the height of these rectangles is at most $\frac{\varepsilon}{4}$. So, and we will just be assuming the max height and that it does not cover a length over 1:

$$< n \cdot l + \sum_{\substack{\text{rectangles not}\\ \text{including } c_i}} b_i \left(\frac{\varepsilon}{4}\right)_{\text{max ht of each rect}} < n \cdot l + (1)_{\text{dist from 0 to 1}} \frac{\varepsilon}{4} = \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}$$